INCH-POUND
MIL-DTL-32154 (OS)
08 September 2004

DETAIL SPECIFICATION
CORD ASSEMBLY, DETONATING, SHIELDED MILD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the department of defense.

1. Scope

1.1 Scope. This specification covers the requirements for the shielded mild detonating cord (SMDC) (part no. 30003-841AS425).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-8500	Interchangeability And Replaceability Of Component Parts For Aerospace Vehicles
MIL-D-21625	Design And Evaluation Of Cartridges For Cartridge Actuated Devices

DEPARTMENT OF DEFENSE HANDBOOKS

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A **FSC 1377**

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MIL-HDBK-728/1 Nondestructive Testing

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

Cage Code	Doc. Number	Title
30003	841AS800	Test Assembly, SMDC
30003	841AS801	Fitting Sensitivity Test
30003	841AS450	Test Tip, Det Cord

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E545	Standard Test Method for Determining Image Quality in Direct Thermal Neutron Radiographic Examination
ASTM E1417	Standard Practice for Liquid Penetrant Examination
ASTM E1444	Standard Practice for Magnetic Particle Examination
ASTM E1742	Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/cgi-bin/SoftCart.exe/STORE/standardsearch.shtml?L+mystore+afpt6301+1062008944 or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4 Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. GENERAL REQUIREMENTS.

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3.1 First article sample. Unless otherwise specified by the contract or purchase order, an acceptable first article sample of 18 SMDC shall be required when more than 12 months have elapsed since the production of the latest accepted production lot. The sample shall be tested in accordance with 4.2.

3.1.1 Design and construction. The SMDC design and construction, excluding bend configurations, shall be identical to that subjected to qualification testing and approved for service use. The cord, end fittings and booster tip shall form a continuous metallic enclosure, i.e. faraday cage, around the explosive elements such that they are protected from the affects of Electrostatic Discharge (ESD).

3.2 Envelope and installation. The dimensions and configuration of the SMDC shall be as specified on DWG 841AS425. All parts having the same vendor's item number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500.

3.3 SMDC tubing. The tubing required to confine the products of detonation shall be made of stainless steel and shall withstand a hydrostatic proof pressure test of 3000 Psig.

3.4 Detonating cord. The detonating cord utilized in SMDC manufacture shall be silver sheathed Hexanitrostilbene (HNS), Type II, Grade A explosive per 841AS425. Maximum explosive lot moisture content at the time of cord manufacture shall not exceed 0.3 percent by weight. Only one lot of detonating cord shall be used in each SMDC production lot. Silver sheathing shall be 99.9% silver.

3.4.1 Detonation velocity. The detonation velocity of the detonating cord shall be 6500 ± 500 meters per second when tested at -65°F and +350°F in accordance with 4.4.2. The detonation wave shall propagate through the entire length of the test specimen.

3.4.2 Core load. The range of the explosive core loading in grains per foot divided by the nominal core load shall not exceed 24 percent when calculated from the detonating cord sample data obtained from 4.4.3.

3.5 Transition charge. The transition charge located between the end of the detonating cord and booster charge shall be Hexanitrostilbene (HNS), Type I, Grade A explosive per 841AS425. Maximum explosive lot moisture content at the time of SMDC manufacture shall not exceed 0.3 percent by weight. Only one manufacturer's lot of explosive shall be utilized in each SMDC production lot.

3.6 Booster charge. The booster charge contained in the SMDC end cup shall be Hexanitrostilbene, Type I, Grade A explosive per 841AS425. Maximum explosive lot moisture content at the time of SMDC manufacture shall not exceed 0.3 percent by weight. Only one lot of explosive shall be utilized in each SMDC production lot. The booster shall contain 1+/- .1 grain of HNS pressed to 32,000 +/- 1,000 psi.

3.7 Leakage. The SMDC end fitting leakage rate shall not exceed 10^{-5} cc/sec of air when tested in accordance with 4.3.5.

3.8 End fitting strength. The end fitting joint between the outer ferrule and the stainless steel tubing shall be capable of withstanding a tensile load of 800 Pounds minimum applied at a rate of 8 ± 2 Pounds per second with the 800 pound load being held for 3.5 ± 0.5 minutes. See 4.6

3.9 Booster tip inspection. All booster tips shall pass length and concentricity checks when tested in accordance with 4.3.3.1.

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3.10 Performance. The SMDC shall be initiated by the output from a SMDC test tip NAVAIR SCD 841AS450 over the temperature range of –65°F to +350°F and shall deliver sufficient energy to propagate a detonation wave in the next ordnance device or perform the necessary work function. SMDC shall be capable of transferring a detonation wave from one SMDC to another across the air gaps of the sensitivity fitting, NAVAIR drawing 841AS801 and the fittings of applicable installation over the temperature range from -65°F to +350°F. The SMDC shall be constructed such that the portion of the assembly, external of the SMDC ports, remains intact when fired without rupturing, splitting cracking, emitting shrapnel, disengaging from end fittings or shearing threads. The minimum depth of indentation produced in 6061 – T6 aluminum block shall be 0.040 inch when the SMDC is fired in accordance with 4.7.

3.11 Sets. Individual SMDC may be procured separately by applicable item dash numbers or sets may be procured by set numbers. Each set shall be composed of the SMDC dash numbers indicated in on the applicable drawing.

3.12 Workmanship. Workmanship shall be in accordance with high grade manufacturing practices to assure safety, proper operation and service life. Assembled units shall be free of mechanical damage as defined in approved manufacturer's acceptance procedure.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article sample. The first article sample selected in accordance with 3.1 shall be sequentially subjected to the following tests of MIL-D-21625 in the order listed: Temperature and Humidity Cycling (28 Days). Salt Spray; Vibration and Shock. After the completion of the sequential environmental treatments, the sample shall be inspected and x-rayed for damage. Any damages inflicted by the sequential environmental treatments which would adversely affect the performance of the item in service application shall be cause for the rejection of the first article sample. The sample shall be divided into two equal groups and fired in accordance with 4.7.2.1 and 4.7.2.2. The sample shall be acceptable with respect to functional testing if each firing meets the requirements of 3.10.

4.3 Nondestructive tests.

4.3.1 Metal parts inspection. All high stress steel parts capable of being magnetized shall be subjected to magnetic particle inspection in accordance with ASTM E1444 as supplemented by an approved contractor process specification. Titanium, aluminum or non-magnetic steel high stress parts shall be inspected in accordance with ASTM E1417 as supplemented by an approved contractor specification. Stainless steel tubing shall be subjected to Eddy current testing in accordance with an approved contractor process specification based on the principles and guidelines of MIL-HDBK-728/1. Discontinuities revealed by nondestructive testing may be removed by reworking. If it is established that such discontinuities cannot be removed by subsequent operations or other suitable mechanical removal techniques, the parts shall be rejected. All parts which have had discontinuities removed by machining shall be re-inspected to assure defect removal.

4.3.2 Tubing hydrostatic pressure test. All stainless steel tubing used in the SMDC shall be subjected to a hydrostatic proof pressure test of 3000 Psig (Minimum) prior to assembly. There shall be no evidence of leakage, yielding, or permanent deformation.

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4.3.3 Dimensional. The SMDC shall be inspected to ensure compliance with 3.2.

4.3.3.1 Booster tip inspection. A 100% inspection of booster tip for length and concentricity shall be performed using gage set (P/N A51S62680 – 1) or an approved equivalent. All defects shall be removed from the lot.

4.3.4 Visual. Each SMDC produced under contract shall be visually inspected. Units shall be rejected which have illegible, missing or inaccurate identification markings, missing o-rings, damaged or insufficient surface coatings, burrs, sharp edges, corrosion, dents, or other defects that might prevent proper installation or performance. As an inspection aid, each SMDC should be visually compared with an inert master configuration SMDC to check the location of identification tags and to verify the proper bend configuration, assembly color, and size of fittings installed. The inert master configuration SMDC shall be verified by virtual analysis of vector data prior to manufacture of each SMDC line. Vector data of the master line will be provided for virtual analysis by the aircraft manufacturer. Each line shape must be approved in writing prior to manufacture.

4.3.5 Leak test. SMDC end fittings shall be leak tested in a dry gas leak tester by the contractor to determine conformance to 3.7. SMDC end fittings which exhibit a leak rate in excess of 10^{-5} cc/sec of air at a pressure differential of one atmosphere shall be considered defective and rejected.

4.3.6 Radiographic inspection. Radiographic inspection plates shall be properly identified by the NAVAIR drawing number and item dash numbers, the complete lot number and serial numbers of the SMDC displayed thereon. SMDC shall conform to the following standard when subjected to the following radiographic examinations of 4.3.6.1 and 4.3.6.2. A maximum of two bright spots, indicating the presence of foreign particles, will be allowed in each booster shell load. No particle shall exceed 0.005 inch in any dimension. No particles are allowed on explosive interfaces of pressed loads. No foreign material is allowed in the ferrule and cord explosive loads. Epoxy located on the shoulder of the booster cup and visible on the radiograph is acceptable, providing the epoxy may be verified by visual examination of the SMDC and does not mask the cord to ferrule interface. An excessive amount of epoxy on the booster cup, sufficient to interfere with the interpretation of the radiograph is not acceptable. The gap between the inner ferrule charge and booster cup charge shall be no more than 0.005 inch over at least a continuous width of 0.100 inch as viewed on the N-ray. A change in N-ray film density at the ferrule/booster cup interface caused by the inner ferrule load not being exactly flat and/or a high inner ferrule load is permissible. No voids are permitted at any other place. Units containing one or more defects shall be rejected.

4.3.6.1 N-Ray examination. Each assembled SMDC shall be N-rayed in accordance with ASTM E545 prior to bending along the full length of the longitudinal axis in serial number sequence and the plates examined for defects in accordance with approved manufacturer's radiographic procedures.

4.3.6.2 X-Ray examination. Each assembled SMDC shall be x-rayed in accordance with ASTM E1742 after bending at each tip and the adjacent three inches of detonating cord along the longitudinal axis in serial number sequence and the plates examined for defects in accordance with approved manufacturer's radiographic procedures.

4.4 Detonating cord tests.

4.4.1 Sampling. Prior to its use in SMDC manufacture, 21-inch long samples shall be selected from each tube in a lot of detonating cord in the following manner. Commencing with the initial 21 inches of detonating cord on the spool or coil, a 21 inch sample shall be taken at each 100 foot interval except for

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the final sample which shall be the last 21 inches at the end of the cord on the spool or coil. Each sample shall be divided into three segments, two 9 inches long and the other 3 inches long, for the following tests. The failure of any sample to meet the requirements of 3.4.1 and 3.4.2 shall result in the rejection of the sampled tube.

4.4.2 Detonation velocity. The 9 inch segments of the detonating cord sample shall be temperature conditioned for not less than 8 hours nor greater than 24 hours, one segment each at -65° ± 5°F and +350° ± 5°F. Following temperature conditioning, each segment shall be fired by a commercial no. 6 electrical blasting cap and the detonation velocity measured to determine conformance to the requirements of 3.4.1.

4.4.3 Core loading. The amount of explosive contained in the 3 inch segment of the detonation cord samples shall be determined to verify conformance to the requirements of 3.4.2

4.5 Explosive lot moisture content. Prior to the loading of explosives into components, the moisture content of the explosives shall be determined to verify conformance to the 0.3 percent maximum moisture content specified in 3.4, 3.5 and 3.6.

4.6 End fitting tensile strength. A tensile strength test sample shall be randomly selected from the SMDC lot prior to bending in accordance with the provisions for normal inspection at inspection level S-1, ANSI/ASQ Z1.4 and the following exception: rejected test specimen from other tests of this source control drawing may be utilized for the tensile strength test provided that the previous rejection was unrelated to the end fitting tensile strength. The sample thus selected shall be tested by the contractor in a tensile strength testing machine to determine conformance with the requirements of 3.8. The acceptance number for defectives shall be 0 and the rejection number shall be 1.

4.7 Functional tests.

4.7.1 Lot samples. Unless otherwise specified, the sampling plans and procedures used in the determination of the acceptability of lot samples submitted by the supplier to the test activity designated in the contract shall be in accordance with ANSI/ASQ Z1.4, inspection level I, Table II-A, normal inspection. The acceptance number for defectives shall be 0 and the rejection number shall be 1. All tests of 4.3 shall have been satisfactorily completed prior to the submittal of the lot sample to the test activity designated in the contract. The sample shall be randomly selected from the SMDC lot prior to bending in a manner that will not produce a negative effect on planned deliverables. Each SMDC selected shall then be bent in the standard SMDC lot acceptance test configuration shown on DWG 841AS425.

4.7.2 Test procedures. The sample selected from the production lot in accordance with 4.7.1 shall be divided, as nearly as possible, into two equal test groups for low and high temperature functional testing. If the sample is not equally divisible by two, the extra test specimen shall be placed in the low temperature group. All test groups shall be temperature conditioned for a period of time not less than 4 hours or greater than 24 hours.

4.7.2.1 Low temperature test (-65°F). The test group selected for the low temperature functional test shall be temperature conditioned at -65°F and fired in accordance with the low temperature functional test requirements of MIL-D-21625. The test specimen shall be fired in the manner depicted on NAVAIR drawing No. 841AS800.

4.7.2.2 High temperature (+350°F). The remaining test group shall be temperature conditioned at +350°F and fired in accordance with the high temperature functional test requirements of MIL-D-21625

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except that the conditioning and firing temperatures shall be +350°F in lieu of +200°F. The test specimen shall be fired in the manner depicted on NAVAIR drawing 841AS800.

4.7.2.3 Acceptance criteria. The lot shall be acceptable with respect to functional testing if the low and high temperature groups meet the requirements of 3.10 at an AQL of 0.65 in accordance with ANSI/ASQ Z1.4.

4.7.2.4 Retest. There shall be no retest.

4.7.2.4.1 Test failure. If a test failure is attributable to an assignable cause, excluding the SMDC test specimen, at the option of the test activity that portion of the test may be re-conducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

6.1 Intended use. SMDC lines are intended as a rapid energy transmission devices to provide and an explosive release signal in aircraft emergency egress systems. The SMDC lines were developed for use in military aircraft and have no commercial application.

6.2 Acquisition requirements. Procurement documents should specify the following:

A. Title, number and revision of this specification, Applicable dash numbers or set numbers of items being procured by this specification.
B. Whether first article inspection is required, and if so specify the test activity.
C. Test activity.
D. Items of data required for each first article and production lot.

6.3. Explanation of terms. For the purpose of this drawing, NAVSUPINST 4030.28B (15 January 1984), when applicable shall be used to clarify terms.

6.4 Subject term (keyword) listing.

Aircraft emergency egress systems
Hexanitrostilbene

6.5 Justification. This specification has been generated by authority of ECP 045220V003.

Preparing activity:
Navy-OS
(Project 1377-0215)

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